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Tectonic rotations south of the Bohemian Massif from palaeomagnetic directions of Permian red beds in Hungary

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Abstract

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Palaeomagnetic studies were carried out in Permian red beds of the Balaton Highlands, the Mecsek Mountains, and the Bükk Mountains of Hungary. Statistically well defined directions were obtained from six localities in the Balaton Highlands and two localities in the Mecsek Mountains. No meaningful results were obtained from the Bükk Mountains.

Three magnetic components were identified from red beds of the Balaton Highlands: (1) in haematite with a very high unblocking temperature (700 °C), interpreted as a Permian magnetization ($D_c = 79°$, $I_c = -11°$, $k = 24$, $\alpha_{95} = 13.6°$), in six samples from three beds in a single locality; (2) a secondary but ancient component residing mainly in maghemite ($D = 314°$, $I = 49°$, $k = 48$, $\alpha_{95} = 10.0°$), in 84 samples from six localities with a within-locality scatter increasing on unfolding; and (3) a direction parallel to the present field ($D = 7°$, $I = 62°$, $k = 46$, $\alpha_{95} = 7.7°$), in nine samples from a single locality. For the Balaton Highlands, the component 1 direction agrees with directions obtained from Permian red beds and volcanics in the eastern part of the Southern and Eastern Alps and the Inner West Carpathians. All show large, apparent rotations relative to stable Europe since the Permian. Component 2 is of post-folding (post-Aptian) age. Its direction agrees with known Late Cretaceous directions from the Transdanubian Central Mountains, which also show significant counterclockwise rotation relative to stable Europe.

The characteristic magnetization for the Mecsek Mountains resides in haematite and may be primary. The directions indicate only a slight net counterclockwise rotation of the Mecsek Mountains with respect to stable Europe since the Permian.

Introduction

During the past decade a variety of previously unstudied rocks of Hungary, mainly weakly magnetized limestones and red, grey and white marls, and migmatitic and intrusive granites, have yielded palaeomagnetic directions that indicate significant past rotations with respect to the present orientation (Márton and Márton, 1978, 1980, 1981, 1983; Márton 1980, 1984a, b). However, the Permian red beds have remained palaeomagnetically enigmatic.

Hungarian red beds were first investigated in the 1960s (Kotásek et al., 1969). Apart from a single locality (Mecsek Mountains, Egédpuszta: $D = 187°$, $I = -18°$, $k = 14$, $\alpha_{95} = 5.8°$, 48 samples), the sampling was not successful. In the late 1970s, P. Márton investigated red siltstones from the Balaton Highlands and the Mecsek Mountains, but obtained no statistically meaningful palaeomagnetic results.

Red sandstones and siltstones were among the first rocks to be studied palaeomagnetically in Europe because of their relative high intensity and

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stability of magnetization. As a result, a large number of palaeodirections considered to represent the Permian magnetic field have been reported for stable Europe and for adjacent regions tectonically related to Hungary. However, Mesozoic and Palaeogene rocks are practically unexplored in stable Europe, and only a few poorly dated palaeomagnetic directions are reported from Africa, the other large plate inferred to have played an important role in the tectonic development of the Mediterranean region. In contrast to the major plates, the main sources of palaeomagnetic data in the Mediterranean have been Mesozoic and Palaeogene carbonates, precluding direct comparison between directions measured from the Mediterranean and the European and African plates.

Thus, for a lack of a better solution, Mediterranean palaeopoles are usually compared with poles for Africa or Europe with the help of a plate tectonic reconstruction (VandenBerg, 1979, Lowrie et al., 1980, Márton and Márton, 1981, 1983). This procedure allows common trends or large relative movements in the Mediterranean to be recognized with respect to the major plates. However, there are shortcomings to the use of the "pseudo" apparent polar wander path (APWP) and work continues to be directed in the stable areas at developing better defined APWPs. Until additional results are reported from the stable areas, studies in structurally rotated terranes are best continued in strata the directions of which can be directly compared with those already reported from the major plates. Because a Permian direction is one of these, we decided to renew our attack on the problem of obtaining reliable directions from the Hungarian red beds.

Sampling and laboratory treatment

Five localities in the Balaton Highlands, part of the Transdanubian Central Mountains, were sampled by us in 1982(Fig. 1, localities 1–5). These included three siltstone outcrops (localities 1–3) stratigraphically immediately below a red/grey colour boundary considered to mark the Permian/Triassic boundary (although the lowest Triassic fauna appear a few tens of meters above the colour change); one siltstone outcrop (locality 4) below a basal conglomerate considered to mark the onset of Triassic sedimentation; and a locality in a fine- to medium- and coarse-grained sandstone quarry (locality 5) at a lower stratigraphic position. Two localities in the Mecsek Mountains (Fig. 1., localities 8 and 9) and one locality in the Bükk Mountains (Fig. 1, locality 10) were also sampled. Two additional localities in the Balaton Highlands were subsequently sampled by Márton in 1984 (Fig. 1, localities 6 and 7).

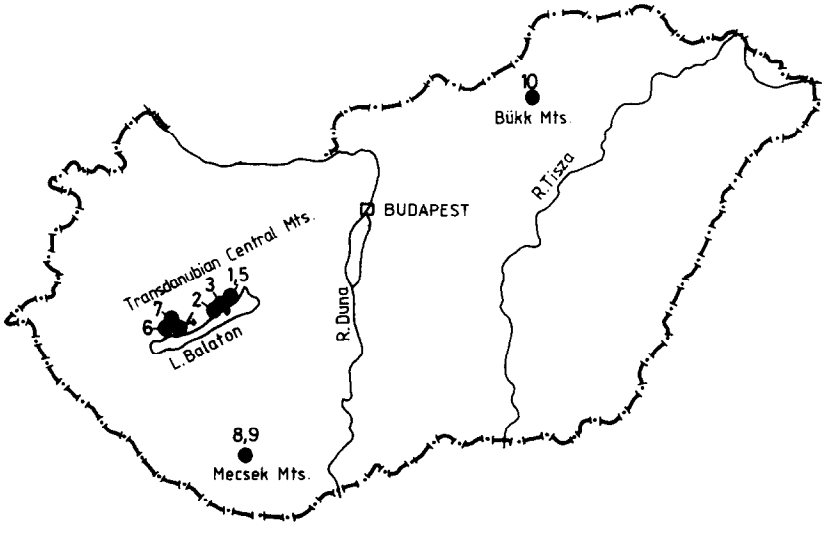


Fig. 1. Sampling localities (sites).

TABLE 1

Permian red beds, Hungary: summary of the palaeomagnetic results

Locality	N/N_o	Palaeomagnetic direction (degrees)				Statistical parameters		Cleaning, remarks
		before tilt correction		after tilt correction		K	α_{95}	
		D	I	D	I			
Balaton Highlands								
1. Balatonalmádi								
47.05° N, 18.02° E	20/20	323	49	319	9	21	7.8	500° C, maghemite decays below 500° C
2. Balatonfüred								
46.95° N, 17.83° E $2a_1$	9/9	308	29	295	0	15	13.8	400° C, maghemite decays below 400° C
$2a_2$	9/9	301	24	293	−8	53	7.1	subtracted vectors 400° −500° C
3. Balatonarács								
46.96° N, 17.87° E	29/29	322	53	307	18	25	4.7	subtracted vectors NRM-525° C
4. Kővágóörs								
46.83° N, 17.62° E	8/12	309	47	310	24	113	4.9	subtracted vectors 400° −500° C maghemite decays at 525° C
6. Badacsonyörs								
46.80° N, 17.47° E 6a	6/13	81	24	79	−11	24	13.6	700° C, maghemite decays below 500° C
6b	11/13	302	47	347	52	34	7.9	subtracted vectors 150° −650° C
7. Salföld								
46.83° N, 17.51° E $7a_1$	9/10	336	63	341	46	124	4.7	300° −475° C maghemite decays below 600° C
$7a_2$	9/10	323	60	332	43	125	4.6	subtracted vectors 300° −575° C
7b	9/10	7	62	1	45	46	7.7	575° C
Mecsek Mountains								
8. Egédpuszta								
46.10° N, 18.00° E	11/11	167	−20	185	−10	35	6.1	525° C, maghemite decays below 525° C
9. Boda								
46.10° N, 18.00° E	17/17	188	−31	170	0	25	9.8	500° C, maghemite decays below 500° C

N_o–number of collected samples; N—number of samples used for statistical evaluation (the rejected samples are either unstable or do not show the component in question); D–declination; I—inclination.

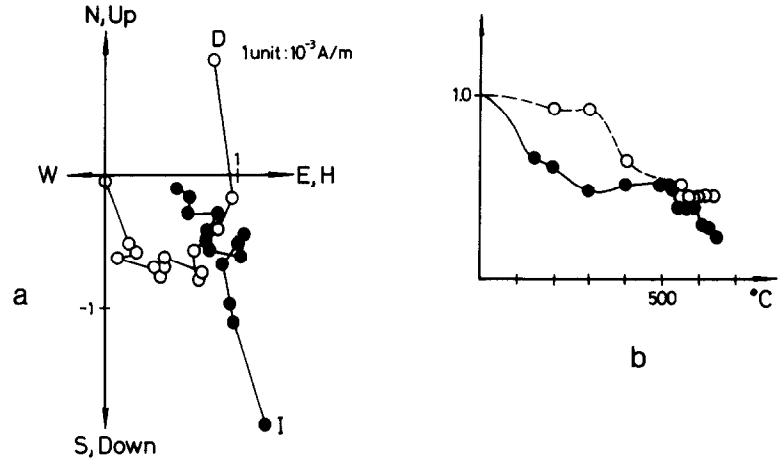


Fig. 2. Mecsek Mountains. Locality 8, sample EP2 1.9. a. Orthogonal plot of the components of the NRM measured at successive cleaning steps. b. Normalized intensity (dots) and susceptibility (circles) values as a function of temperature.

The 1982 collection was processed in Flagstaff, Arizona. Following the measurement of natural remanent magnetization (NRM), pilot specimens were thermally demagnetized stepwise, with susceptibility as well as remanence measured after each heating step (Fig. 2). Based on the behaviour of the pilot samples, the entire collection was cleaned at selected temperatures, with selection at first based on the disappearance of maghemite. However, it soon became clear that for some samples only maghemite carried a statistically well defined direction (e.g., Table 1, locality 3). The collections were thus partially demagnetized before as well as after the susceptibility decrease observed in the pilot specimens. In Flagstaff, an ScT cryogenic magnetometer was used to measure both the remanence and the susceptibility, and a home-made oven was used for thermal demagnetization analysis.

The samples collected in 1984 were processed in the palaeomagnetics laboratory of the Eötvös L. Geophysical Institute, Budapest, Hungary, employing a different measuring and evaluation system, and all samples were treated as pilot samples. The remanence was measured on JR-4 magnetometers, the susceptibility on a KLY-2 bridge, and the oven was a Schoenstedt TSD-1. One specimen per sample was heated stepwise, and the susceptibility as well as the remanence were measured after each step (Fig. 3).

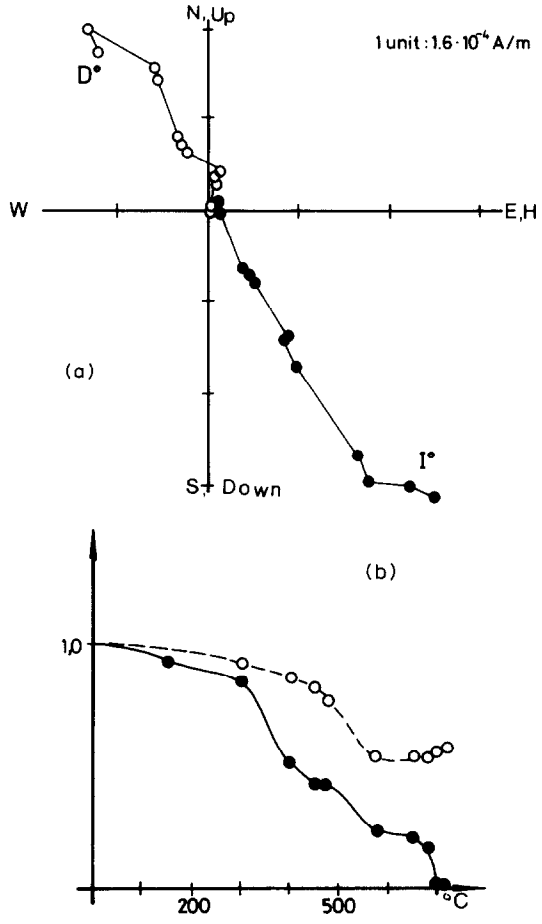


Fig. 3. Balaton Highlands. Locality 6, sample P 302. a. Orthogonal plot of the components of the NRM measured at successive heating steps. b. Normalized intensity (dots) and susceptibility (circles) values as function of temperature.

Results

Márton and Elston (1985) reported the laboratory experiments for localities 1–5, and 8–10, most of which yielded statistically well defined directions (localities 5 and 10 were rejected because of the large statistical scatter at every stage of the cleaning).

In each case, the direction of the characteristic remanence departed significantly from the present field direction (Table 1, localities 1–4 and 8–9, *D* and *I* before tilt correction), indicating a high stability of the remanence in time, although the main carrier of magnetization was identified as maghemite at localities 1–4. The inclinations became sufficiently low after tilt correction to correspond to a Permian magnetization. However, unfolding had no effect on the between-locality scatter, i.e. the age of the statistically well defined stable directions of magnetization remained an open question.

One problem with attempting a fold test for Permian red beds initially sampled in the Balaton Highlands was a similarity of tectonic attitude (bedding corrections recorded at localities 1–4 were as follows: 1, 290/26; 2, 240/56; 3, 295/40; 4, 314/23). However, localities sampled in 1984 came from strata with different tilts (6, 62/36; 7, 350/18) and for this reason they play an important role in the interpretation of the timing of the magnetization.

The NRM is multicomponent in samples from localities 6 and 7. At locality 7, a component isolated at around 575 °C parallels the present field direction (Table 1, 7b). At lower temperatures, another component is seen both in the measured vectors ($7a_1$) and in the subtracted vectors ($7a_2$). Both components (7a and 7b) seem to reside in maghemite. The component with lower unblocking temperature appears to be slightly biased towards the present field direction.

Samples from locality 6 have an overprint component (Table 1, 6b) partly residing in maghemite which is very similar in direction to 7a. The characteristic magnetization was isolated at 700 °C in six samples after exhaustive cleaning (Figs. 4 and 5), and it differs from components observed in all other Permian red beds of the Balaton Highlands.

The results of the laboratory analysis of the NRM of the samples from the Balaton Highlands may be summarized as follows:

(1) Complete overprint in the present field was not observed.

(2) With the exception of 6a (Table 1) the

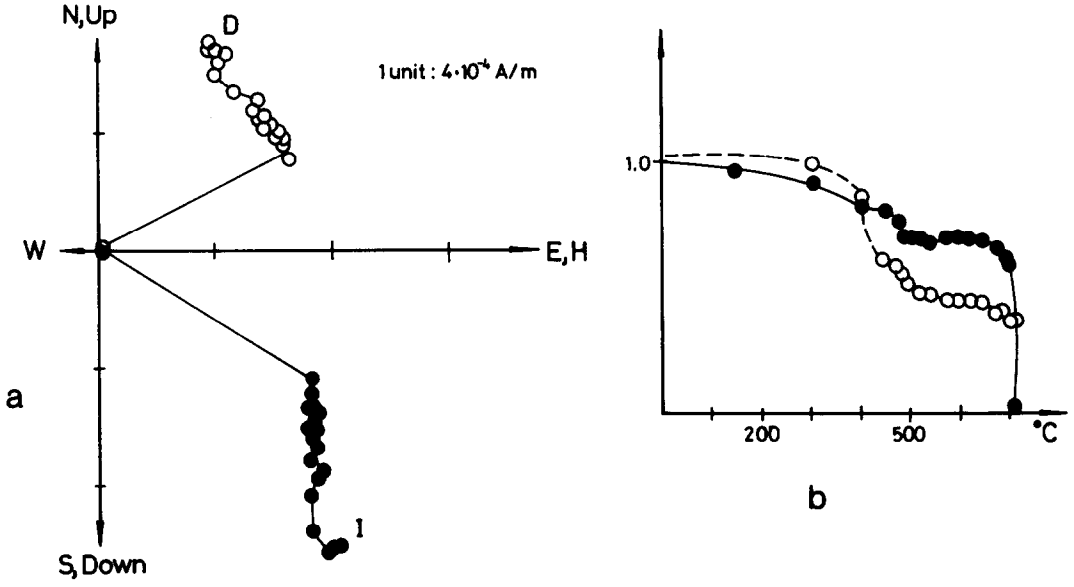


Fig. 4. Balaton Highlands. Locality 6, sample P 303. a. Orthogonal plot of the components of the NRM measured at successive cleaning steps. The continuous removal of the overprint is indicated by the smooth D and I paths. b. Normalized intensity (dots) and susceptibility (circles) values are shown as a function of cleaning temperature.

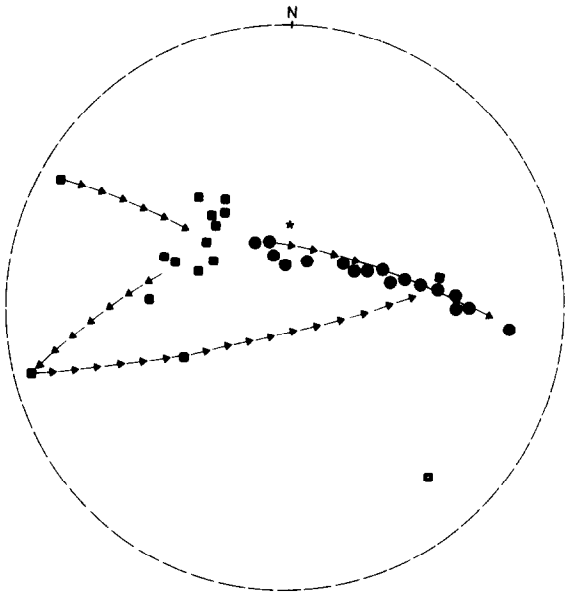


Fig. 5. Balaton Highlands. Locality 6, sample P 303. Stereographic plot of the directions of the measured (circles) and removed (squares) remanence. Closed symbols—lower hemisphere; open symbols—upper hemisphere. For cleaning steps refer to Fig. 4b.

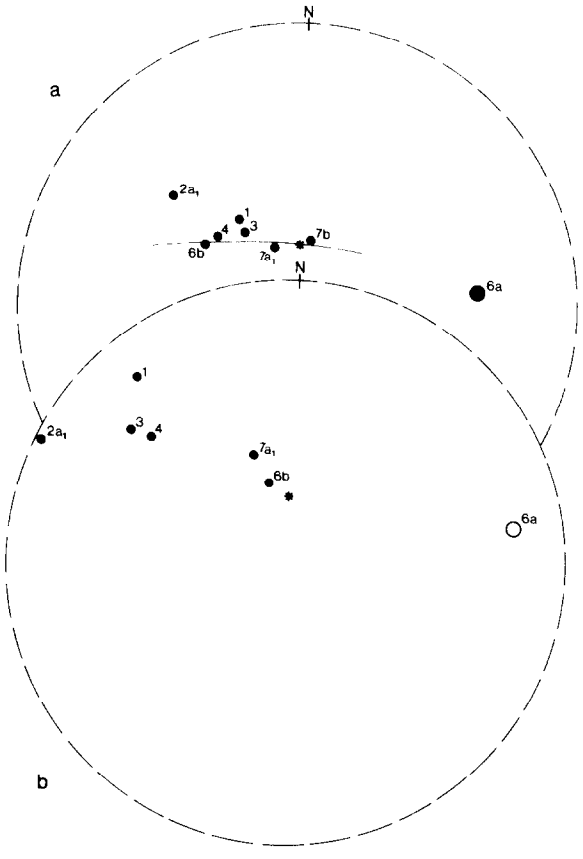


Fig. 6. Balaton Highlands. Locality means before (a) and after (b) tilt correction. Stereographic plot. Dot—positive inclination; circle—negative inclination; asterisks—direction of the present local magnetic field.

palaeomagnetic components of both characteristic magnetizations (ChRM) and components with a lower unblocking temperature are aligned along a great circle defined by the present field direction and a counterclockwise rotated direction that plots close to 6b, 4, 1 and 3 (Fig. 6(a)); the same components are connected to maghemite, except 7b which is carried by haematite. The "streaking" suggests that the formation of the maghemite was a slow process, occurring over a geologically appreciable time span and mainly in a field different from the present.

(3) Component 6a (Fig. 6(a)) departs significantly from the present field direction as well as from the cluster of the other directions; it is connected to haematite which is extremely resistant to thermal cleaning.

The previously developed data set shows that the NRM is less complicated in the studied red beds from the Mecsek Mountains; the characteristic magnetization at both localities is carried by haematite and differs significantly from the present field direction and from any direction observed in the red beds from the Balaton Highlands (Table 1, localities 8 and 9).

Discussion

The complex NRM at locality 6 (Table 1) serves as a base for evaluating the timing of acquisition of the remanence of red beds from the Balaton Highlands. The component 6a direction is considered to be a primary remanence, although not necessarily of depositional origin for several reasons. It resides in haematite that is extremely resistant to thermal cleaning; the structurally uncorrected direction differs significantly from the present field direction, and the tilt-corrected inclination is low enough to be a Permian inclination. Additionally, the polarity may be concluded

to be reverse by correlation with the already proven post-Permian counterclockwise rotation of the Transdanubian Central Mountains (e.g. Márton and Márton, 1983). Lastly, the declination after tilt correction agrees with some other directions interpreted as Permian from the eastern part of the Southern and Eastern Alps and the Inner West Carpathians (Fig. 7).

The overprint component at locality 6 (Table 1, direction 6b), although resistant to heating and differing significantly from the present field direction, is less resistant to thermal cleaning than direction 6a. It shows a remarkable similarity to directions observed at other localities where the magnetization resides solely in maghemite. The scatter of directions 6b, 7a, 3, and 4 increases on

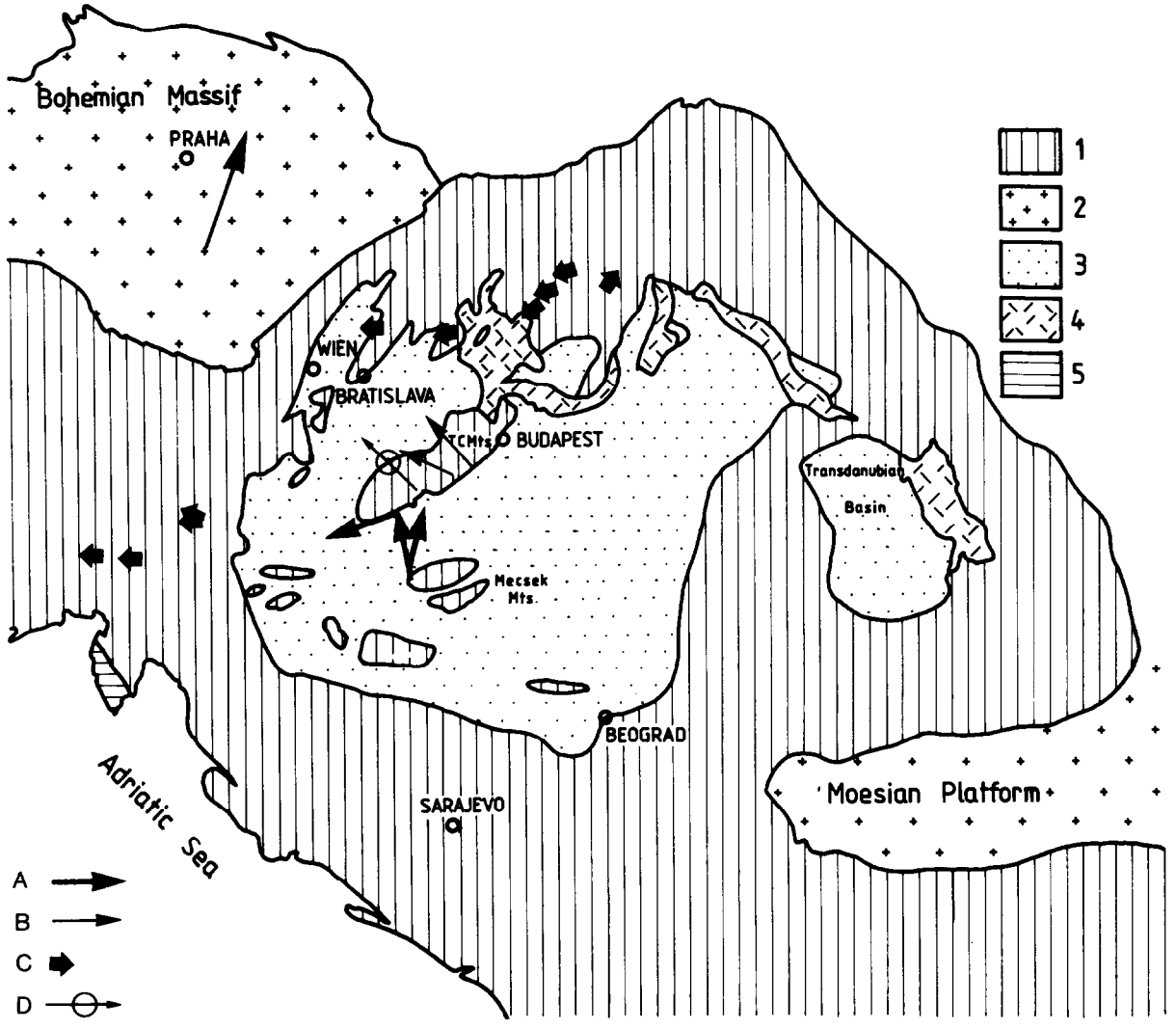


Fig. 7. Palaeomagnetic declinations for Permian rocks of the Balaton Highlands and the Mecsek Mountains compared with directions for Permian rocks of the Southern and Eastern Alps (Mauritsch and Becke, 1983; Mauritsch, pers. commun., 1985) and the Inner West Carpathians (Krs et al., 1982). The mean Permian declination for stable Europe is calculated on the basis of the following determinations: average for the Bohemian Massif (Krs, 1969), Thüringian Forest (Mauritsch and Rother, 1983), Black Forest (Konrad and Nairn, 1972), Sauerland and Harz (Bachtadse, 1983). Key to geology: *1* = area of Alpine folding; *2* = crystalline basement; *3* = basin sediments; *4* = Tertiary volcanic area; *5* = stable foreland. Key to the directions (arrows show the direction of the palaeodeclination): *A* = directions interpreted as primary form the Transdanubian Central Mountains (6a, Table 1) and the Mecsek Mountains (8 and 9, Table 1); *B* = possibly Permian directions from the Transdanubian Central Mountains (1 and 2, Table 1); *C* = directions interpreted as Permian from the Southern and Eastern Alps and the Inner West Carpathians; *D* = postfolding magnetization of the Permian rocks from the Balaton Highlands.

tilt correction, suggesting a post-tilting age for this component of magnetization. Although this remanence clearly is secondary, nonetheless it is ancient.

The characteristic remanences at localities 1 and 2 are difficult to evaluate. If a primary age is assumed at these sites, we are confronted with the problem of explaining an early destruction of a "primary" ChRM carried by specularite and an associated early haematite pigment. On the other hand, the structurally uncorrected mean direction of locality 2 plots away from the cluster of other secondary directions, and agreement of the structurally uncorrected ChRM direction of locality 1 with the secondary directions may be a coincidence.

In any case, a general interpretation of palaeomagnetic results from the Balaton Highlands in terms of tectonics is basically independent of the correct answer to the above question. With respect to stable Europe, represented by the Bohemian Massif in the present vicinity of the Pannonian Basin, a large counterclockwise rotation of the Balaton Highlands is evident. The amount of rotation exceeds 90° for the direction interpreted as Permian, and appears distinctly less for the overprinted normal polarity directions. The age of the first acquisition of an overprint magnetization carried by maghemite may be estimated as mid-Cretaceous on palaeomagnetic grounds (Fig. 8), i.e. it ties up with the geologically recognized uplift of the Transdanubian Central Mountains.

The Permian rocks from the Mecsek Mountains yielded palaeomagnetic directions that are close to the Permian direction for stable Europe, although slightly but distinctly rotated counterclockwise relative to it.

The conclusion that the Permian direction from

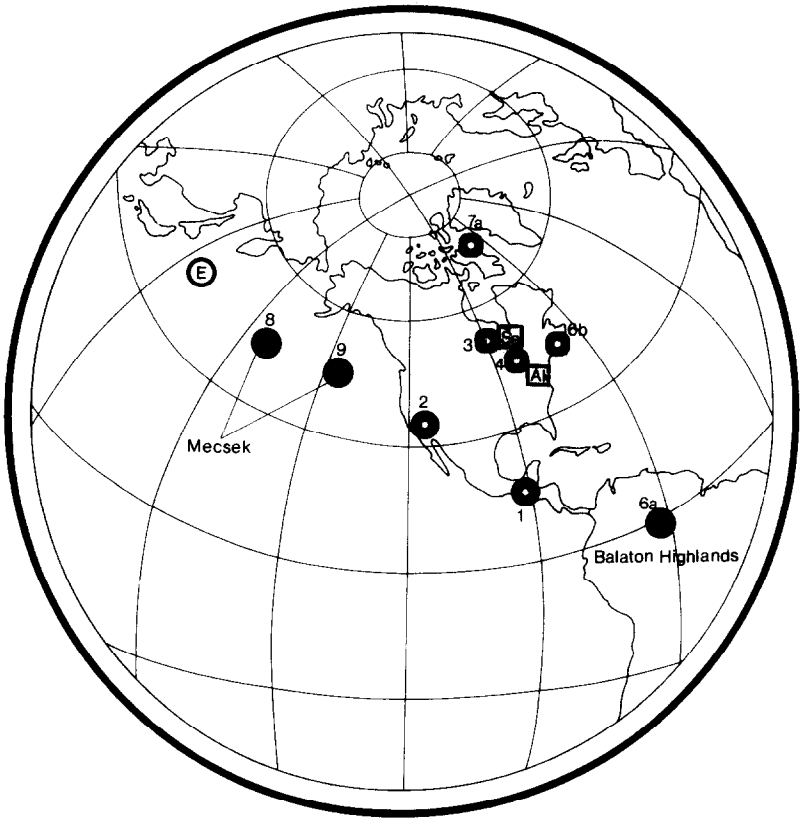


Fig. 8. Palaeomagnetic poles determined from Permian red beds from Hungary (6a, 8 and 9 interpreted as Permian magnetizations, 1 and 2 as possibly Permian magnetization, 3, 4, 6b as magnetization of postfolding age) compared with an average Permian pole for stable Europe (*E*)—based on the determinations as given in the caption of Fig. 7—and Cretaceous poles for the Transdanubian Central Mountains. *Se*—Senonian pole; *Al*—Albian pole.

the Mecsek Mountains is rotated with respect to stable Europe is different from that of Kotasek et al. (1969), although the mean directions for the same locality (Table 1, locality 8) determined by Kotasek et al. (1969) and by us (Márton and Elston, 1985) are similar. The recognition of the moderate counterclockwise net rotation of the Mecsek since the Permian is due to the improved statistics for locality 8 and an additional successful locality (Table 1, locality 9).

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